

Supplementary Material for When Is the Twisted Signing of an Even Cycle Square Spectrally Optimal?

Anonymous

Abstract

This supplement contains a structural refinement of the interface theory, the complete single-gap witness data, and the reproducibility record for the computer-assisted portions of the article. The separated-interface cluster theorem gives the exact $2r$ -eigenvalue count near c_6 , but is not used in the classification proof. The final section distinguishes producer code, independently reconstructed objects, and shared symbolic routines.

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1 Exact $2r$ clusters for separated interfaces

1.1 Operators and the one-interface input

Let $Q^{(6)} \in \{\pm 1\}^{\mathbb{Z}}$ be the flux word whose positive sites are

$$\{-4j : j \geq 0\} \cup \{6 + 4j : j \geq 0\}.$$

Choose a lift $\tau^{(6)}$ with $\tau_{i+1}^{(6)} = Q_i^{(6)} \tau_i^{(6)}$, and define

$$(A_6 u)_i = u_{i-1} + u_{i+1} + \tau_{i-2}^{(6)} u_{i-2} + \tau_i^{(6)} u_{i+2}, \quad H_6 = A_6^2. \quad (1)$$

The other lift is $-\tau^{(6)}$; after squaring it is diagonally unitarily equivalent to the displayed operator. Reflection gives the reverse orientation.

Define

$$\begin{aligned} p_6(y) = & 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 - 423904y^5 \\ & + 484528y^4 - 270464y^3 + 137856y^2 - 19968y + 256. \end{aligned} \quad (2)$$

The constant c_6 is the unique root of p_6 in the rational interval

$$\frac{7905369311620327}{10^{15}} < c_6 < \frac{7905369311620328}{10^{15}}. \quad (3)$$

The period-eight reference squared edge is

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}} < c_6. \quad (4)$$

The proof below uses the following certified one-interface and Floquet inputs. They are the only computer-assisted spectral inputs to this section:

$$\ker(H_6 - c_6) = \text{span}\{\psi_+, \psi_-\}, \quad \dim \ker(H_6 - c_6) = 2, \quad (5)$$

$$H_6|_{\ker(H_6 - c_6)^\perp} \leq c_6 - \frac{1}{100}, \quad (6)$$

where the modes are orthonormal and satisfy $A_6 \psi_\pm = \pm \sqrt{c_6} \psi_\pm$. For all eight cuts of the period-eight bulk, both stable multipliers have modulus below

$$q_F = \frac{9}{25}, \quad (7)$$

and the selected two-column Floquet bases have condition number below 17. These bounds hold for both tails and are uniform under reflection.

1.2 Finite-ring geometry and patch identification

Work on $\mathbb{Z}/n\mathbb{Z}$ in Hamilton gauge. All step-one signs are positive except for a movable seam carrying a holonomy $\alpha \in \{+1, -1\}$. The flux word is $Q_i = \tau_i \tau_{i+1}$. Assume that its positive sites have cyclic gaps four except at exactly $r \in \{1, 2, 3\}$ marked gaps of length six. Write p_j for the left positive endpoint of the j -th marked gap, ordered cyclically, and put

$$d_j = p_{j+1} - p_j \pmod{n}, \quad D = \min_j d_j, \quad (8)$$

with $p_{r+1} = p_1$. For $r = 1$, the return arc has length $d_1 = D = n$. Thus D is a distance in sites, not a number of period-eight cells. Assume from now on that

$$D \geq 1040, \quad S = \left\lfloor \frac{D}{4} \right\rfloor, \quad a = S - 8. \quad (9)$$

On the oriented arc from p_j to p_{j+1} , use the coordinate $0 \leq t \leq d_j$. Define interface cutoffs χ_j and a bulk cutoff β_j by

range of t	χ_j	β_j	$\vartheta = \frac{\pi(t-a)}{2S},$
$0 \leq t \leq a$	1	0	
$a < t < a + S$	$\cos \vartheta$	$\sin \vartheta$	
$a + S \leq t \leq d_j - a - S$	0	1	

(10)

and on the right transition by

$$\beta_j = \cos \varphi, \quad \chi_{j+1} = \sin \varphi, \quad \varphi = \frac{\pi(t - d_j + a + S)}{2S} \quad (11)$$

for $d_j - a - S < t < d_j - a$, followed by $\chi_{j+1} = 1$ on $d_j - a \leq t \leq d_j$. Endpoint values agree. For $r = 1$, the two endpoint pieces are the same cyclic interface cutoff. Pointwise,

$$\sum_{j=1}^r \chi_j^2 + \sum_{j=1}^r \beta_j^2 = 1. \quad (12)$$

The pure-bulk plateau on the j -th arc is $[p_j + 2S - 8, p_{j+1} - 2S + 8]$, and its length is at least

$$d_j - 4S + 16 \geq D - 4\lfloor D/4 \rfloor + 16 \geq 16. \quad (13)$$

For a set E , let $E^{[k]}$ denote its cyclic k -neighborhood. Set

$$I_j = \{i : \chi_j(i) \neq 0\}, \quad J_j = I_j^{[4]}, \quad B_j = \{i : \beta_j(i) \neq 0\}, \quad C_j = B_j^{[4]}.$$

Since $H = A^2$ has range four, the coefficient collar needed to identify a compression to J_j is contained in $I_j^{[6]}$. That collar stays at distance at least

$$D - 2\lfloor D/4 \rfloor - 4 \geq 516 \quad (14)$$

from every other marked core. Likewise $B_j^{[6]}$ stays at distance at least $S - 19 \geq 241$ from each six-gap core. Hence every interface patch contains exactly one marked core and every bulk patch is a pure period-eight patch.

The holonomy seam can be moved by diagonal switching. Place it on an edge of a bulk plateau whose endpoints have distance at least eight from every interface support. It then misses every $I_j^{[6]}$. Each interface collar is a proper arc, so after unwrapping it, translation, optional reflection, and diagonal switching identify its coefficients exactly with the corresponding finite section of H_6 . On a proper arc, two lifts of the same Q -word differ by a global sign. If $(D_0 u)_i = (-1)^i u_i$, then

$$A_{-\tau} = -D_0 A_\tau D_0, \quad H_{-\tau} = D_0 H_\tau D_0. \quad (15)$$

This removes the remaining lift choice after squaring. The same argument identifies each C_j with a finite section of the period-eight reference operator. We record the result in the form needed below.

Lemma 1.1 (Patch identification). *For every allowed orientation, lift, and holonomy, there are local unitaries such that:*

- (i) *the compression of H to J_j is a principal finite section of the corresponding infinite $G6$ operator;*
- (ii) *the image under H of a vector supported in I_j is supported in J_j and agrees with the image under that infinite operator;*
- (iii) *the analogous compression and map on $B_j \subset C_j$ agree with a period-eight reference-bulk finite section.*

These statements include cyclic wraparound because every collar is first unwrapped as a proper arc.

Transport the orthonormal pair ψ_+, ψ_- through the same local unitary, before truncation, and call the transported modes $\psi_{j,+}, \psi_{j,-}$. For every finite-ring vector x , zero extension on the unwrapped arc gives

$$\langle \psi_{j,\pm}, \chi_j x \rangle_{\ell^2(\mathbb{Z})} = \langle \chi_j \psi_{j,\pm}, x \rangle_{\mathbb{C}^n}. \quad (16)$$

This identity is the bridge that permits the complete rank-two complement bound (6) to be used on each localized vector.

1.3 Tails, Gram matrix, and the lower count

The cutoff starts changing at distance $S - 8$ from a marked core. Removing the propagation range four leaves

$$L_{\text{site}} = S - 12, \quad \ell = \left\lfloor \frac{L_{\text{site}}}{8} \right\rfloor. \quad (17)$$

The eight site residues, the two tails, the condition-number bound, and the geometric series imply, for either normalized local mode,

$$\|\psi_{j,\pm}\|_{\text{tail}}^2 \leq \frac{16 \cdot 17^2}{1 - q_F^2} q_F^{2\ell} = \frac{10625}{2} q_F^{2\ell} < 73^2 q_F^{2\ell}. \quad (18)$$

Put

$$\phi_{j,\pm} = \chi_j \psi_{j,\pm}, \quad \Phi = (\phi_{1,+}, \phi_{1,-}, \dots, \phi_{r,+}, \phi_{r,-}), \quad G = \Phi^* \Phi. \quad (19)$$

The full modes at one interface are orthogonal. Their truncated cross term is not discarded; rather,

$$|\langle \phi_{j,+}, \phi_{j,-} \rangle| \leq 73^2 q_F^{2\ell}.$$

The same estimate holds between different interfaces because both vectors are then in their tails. With $m = 2r$,

$$\|G - I_m\| \leq m 73^2 q_F^{2\ell}. \quad (20)$$

The hypotheses give $S \geq 260$, $L_{\text{site}} \geq 248$, and $\ell \geq 31$. Since $m \leq 6$,

$$6 \cdot 73^2 \left(\frac{9}{25} \right)^{62} < \frac{1}{2}. \quad (21)$$

Thus the $2r$ columns are independent and $\|G^{-1}\| \leq 2$.

The row-sum bound gives $\|H\| \leq 16$, while $c_6 < 8$. Cutting an exact mode only in its tails and using (18) yields

$$\|(H - c_6)\phi_{j,\pm}\| \leq (16 + 8) 73 q_F^\ell = 1752 q_F^\ell. \quad (22)$$

Define the isometry

$$U = \Phi G^{-1/2} : \mathbb{C}^{2r} \longrightarrow \mathbb{C}^n. \quad (23)$$

Then

$$\|(H - c_6)U\| < 3504rq_F^\ell < \frac{1}{400}. \quad (24)$$

Let E_W be the spectral projection of H onto

$$W = \left[c_6 - \frac{1}{400}, c_6 + \frac{1}{400} \right]. \quad (25)$$

If $\text{rank } E_W < 2r$, a nonzero vector in $\text{ran } U$ is orthogonal to $\text{ran } E_W$. The spectral theorem then makes its residual at least $1/400$, contradicting (24). Therefore

$$\text{rank } E_W \geq 2r. \quad (26)$$

1.4 The codimension- $2r$ complement

Let $V = \text{ran } \Phi$. If $x \perp V$, then (16) gives

$$\langle \psi_{j,+}, \chi_j x \rangle = \langle \psi_{j,-}, \chi_j x \rangle = 0$$

for every interface. Thus each interface-localized vector is orthogonal to the entire two-dimensional space in (5), and (6) applies. Each bulk-localized vector satisfies the stronger upper bound $\eta < c_6 - 1/100$.

The cutoff vector follows a unit-circle quarter arc at angular speed $\pi/(2S)$. If two sites have cyclic distance $h \leq 4$, the chord bound gives

$$\sum_{\gamma} |\gamma(u) - \gamma(v)|^2 \leq \frac{\pi^2 h^2}{4S^2},$$

where γ ranges over all cutoffs. The exact double-commutator IMS identity, the range-four row-sum bound, and $\pi^2 < 10$ consequently give

$$\|E_{\text{IMS}}\| < \frac{320}{S^2} \leq \frac{320}{260^2} = \frac{4}{845}. \quad (27)$$

Using (12) and summing the localized quadratic-form bounds, we obtain on $V^\perp$

$$H|_{V^\perp} \leq c_6 - \frac{1}{100} + \frac{4}{845} = c_6 - \frac{89}{16900} < c_6 - \frac{1}{200}. \quad (28)$$

By the codimension form of min-max, at most $\dim V = 2r$ eigenvalues lie above $c_6 - 1/200$. Since the entire window W lies above this level, (26) sharpens to

$$\text{rank } E_W = 2r. \quad (29)$$

The equality counts algebraic multiplicity and makes no assertion of individual simplicity.

1.5 The $2r$ -dimensional Feshbach equation

Let

$$P = UU^*, \quad Q_\perp = I - P, \quad E = (H - c_6)\Phi. \quad (30)$$

Because $\text{ran } U = V$, estimate (28) holds on $\text{ran } Q_\perp$. If $|z - c_6| \leq 1/400$, then

$$\|(Q_\perp H Q_\perp - z)^{-1}\| \leq 400 \quad \text{on } \text{ran } Q_\perp. \quad (31)$$

Block Gaussian elimination with respect to $P + Q_\perp = I$ gives the effective operator on $\mathbb{C}^{2r}$,

$$H_{\text{eff}}(z) = U^* H U - U^* H Q_\perp (Q_\perp H Q_\perp - z)^{-1} Q_\perp H U, \quad (32)$$

and the finite-ring eigenvalue equation in W is exactly

$$\det (H_{\text{eff}}(z) - z I_{2r}) = 0. \quad (33)$$

Using $U = \Phi G^{-1/2}$ and $Q_\perp H U = Q_\perp E G^{-1/2}$, one obtains

$$\begin{aligned} H_{\text{eff}}(z) - c_6 I_{2r} &= G^{-1/2} \Phi^* E G^{-1/2} \\ &\quad - G^{-1/2} E^* Q_\perp (Q_\perp H Q_\perp - z)^{-1} Q_\perp E G^{-1/2}. \end{aligned} \quad (34)$$

No orthogonality of the uncorrected columns of Φ is assumed. Writing the right side as $T_1 + R_2(z)$, the Gram and residual bounds give

$$\|T_1\| \leq 3504 r q_{\mathbb{F}}^\ell, \quad \|R_2(z)\| \leq 400 r 3504^2 q_{\mathbb{F}}^{2\ell}. \quad (35)$$

Since

$$400 \cdot 3504^2 \left(\frac{9}{25}\right)^{31} < 1, \quad (36)$$

we have $\|R_2(z)\| < r q_{\mathbb{F}}^\ell$. Every root of (33) in W therefore satisfies

$$|\lambda_j - c_6| < 3505 r \left(\frac{9}{25}\right)^\ell, \quad 1 \leq j \leq 2r. \quad (37)$$

Theorem 1.2 (Separated-interface cluster). *Let an admissible finite ring contain exactly $r \in \{1, 2, 3\}$ elementary G6 interfaces, with arbitrary interface orientations, either lift, and either Hamilton holonomy. If their minimum cyclic site separation satisfies $D \geq 1040$, define S, L_{site}, ℓ by (9) and (17). Then*

$$\text{rank } \mathbf{1}_{[c_6-1/400, c_6+1/400]}(H) = 2r.$$

The $2r$ eigenvalues in this window, counted with multiplicity, satisfy (37), and their exact finite-ring equation is the $2r$ -dimensional determinant (33).

1.6 A sufficient exponential onset

The explicit constants also give a continuous sufficient onset for the exponential construction. For every even n ,

$$\rho_-(n)^2 > 8 - \frac{200}{n^2}.$$

For the nonzero residues put

$$B(n) = 8 - \frac{200}{n^2} - \frac{9}{100}.$$

If the corresponding residue construction has r interfaces and minimum separation $D \geq 1040$, the cluster bound gives

$$\rho(A_n)^2 \leq c_6^{\text{up}} + 3505 r \left(\frac{9}{25}\right)^\ell, \quad \ell = \left\lfloor \frac{\lfloor D/4 \rfloor - 12}{8} \right\rfloor,$$

where $c_6^{\text{up}} = 7905369311620328/10^{15}$. The first distance-eligible endpoints are

$n \bmod 8$	n	r	D	ℓ
2	1042	1	1042	31
4	2084	2	1042	31
6	3126	3	1042	31

and exact rational evaluation gives

$$B(n) - \left[c_6^{\text{up}} + 3505r \left(\frac{9}{25} \right)^{31} \right] > \frac{1}{250}$$

in all three rows. Along each residue subsequence, D and ℓ do not decrease, the spectral cap does not increase, and $B(n)$ strictly increases.

For residue zero, the period-eight cap is $1561/200$. At $n = 3120$,

$$8 - \frac{200}{3120^2} = \frac{389375}{48672}, \quad \frac{389375}{48672} - \frac{1561}{200} = \frac{237251}{1216800} > 0.$$

The first four even orders at or above 3120 are 3120, 3122, 3124, 3126, one in each residue class. The preceding endpoint checks and residue-wise monotonicity therefore prove a strict counterexample for every even $n \geq 3120$. We record

$$N_{\text{exp}} = 3120$$

as a sufficient, nonoptimal onset for this exponential estimate. It is not the sharp onset: the finite bridge plus IMS theorem in the main paper gives the stronger value 48.

The restrictions $r \leq 3$ and $D \geq 1040$ enter, respectively, in the uniform Gram estimate and in the seam-clearance, tail, and IMS bounds. The argument proves neither individual simplicity nor a nonzero leading interaction coefficient.

2 Complete integer witnesses for abnormal single gaps

2.1 Canonical model and comparison threshold

For an integer $g \geq 1$, set

$$D_g = \{-4j : j \geq 0\} \cup \{0, g\} \cup \{g + 4j : j \geq 0\}.$$

Let $Q_i = +1$ for $i \in D_g$ and $Q_i = -1$ otherwise. Choose either lift $\tau_{i+1} = Q_i \tau_i$, and define on $\ell^2(\mathbb{Z})$

$$(A_g v)_k = v_{k-1} + v_{k+1} + \tau_{k-2} v_{k-2} + \tau_k v_{k+2}, \quad H_g = A_g^2. \quad (38)$$

Gap four is the unperturbed period-eight bulk. Gap six is the elementary interface of section 1.

For a finitely supported integer vector v , extend it by zero. Since A_g is self-adjoint,

$$\frac{\langle v, H_g v \rangle}{\langle v, v \rangle} = \frac{\|A_g v\|^2}{\|v\|^2}. \quad (39)$$

The certified upper endpoint in (3) shows that it is enough to exceed the rational threshold

$$T = \frac{7905369311620328}{10^{15}} + \frac{1}{250} = \frac{988671163952541}{125000000000000}. \quad (40)$$

For integers $N, D > 0$, define the exact cross-multiplication

$$\Delta(N, D) = 125000000000000N - 988671163952541D. \quad (41)$$

Then $N/D > T$ if and only if $\Delta(N, D) > 0$.

2.2 Six small-gap witnesses

Coordinates in each vector are listed in increasing order on the interval I . The displayed image is listed on $J = I + [-2, 2]$. Every entry follows directly from the four-term formula (38).

Gap $g = 1$. On $I = [-2, 3]$, take

$$\begin{aligned} v &= (2, 0, 4, 4, 6, 5), \\ A_1 v &= (-2, 2, 4, 2, 12, 15, 13, 10, 11, -5). \end{aligned}$$

Then $D = \|v\|^2 = 97$ and $N = \|A_1 v\|^2 = 812$.

Gap $g = 2$. On $I = [-4, 6]$, take

$$\begin{aligned} v &= (1, -1, -2, -1, -4, -5, -6, -2, 1, 4, 2), \\ A_2 v &= (-1, 0, 1, 0, -7, 0, -14, -11, -12, -14, 10, 5, 5, -2, 2). \end{aligned}$$

Then $D = 109$ and $N = 866$.

Gap $g = 3$. On $I = [-5, 8]$, take

$$\begin{aligned} v &= (0, 1, -1, -3, 0, -6, -8, -6, -10, -6, -8, -6, 1, -3), \\ A_3 v &= (0, -1, 0, 2, -2, -8, 0, -17, -22, -18, -28, -18, -23, -16, \\ &\quad -1, -5, -4, 3). \end{aligned}$$

Then $D = 393$ and $N = 3114$.

Gap $g = 5$. On $I = [-2, 7]$, take

$$\begin{aligned} v &= (2, 0, 4, 4, 4, 4, 1, 3, 3, 3), \\ A_5 v &= (-2, 2, 4, 2, 10, 12, 11, 12, 0, 11, 5, 6, 6, -3). \end{aligned}$$

Then $D = 96$ and $N = 764$.

Gap $g = 7$. On $I = [-2, 9]$, take

$$\begin{aligned} v &= (2, 0, 3, 4, 4, 4, 1, 3, 2, 3, 2, 3), \\ A_7 v &= (-2, 2, 3, 1, 10, 11, 10, 12, 1, 10, 3, 10, 4, 5, 5, -3). \end{aligned}$$

Then $D = 97$ and $N = 768$.

Gap $g = 8$. On $I = [-8, 16]$, take

$$\begin{aligned} v &= (4, 4, 4, 3, -3, 3, 9, 1, 19, 22, 21, 22, 4, 16, 8, 12, 5, 6, 0, -4, \\ &\quad -1, 1, 0, 1, 0), \\ A_8 v &= (4, 0, 8, 11, 14, 8, -7, 8, 26, 3, 53, 61, 59, 63, 9, 46, 19, 35, \\ &\quad 10, 21, -4, -8, -3, 4, 1, 1, 1, 1, 0). \end{aligned}$$

Then $D = 2487$ and $N = 19672$.

All exact comparisons for these six witnesses are collected in table 1. The third column is the raw integer from (41); the last column is the reduced or partially reduced exact difference from T .

Every integer and every rational number in table 1 is positive. Thus (39) proves $\sup \text{spec}(H_g) > c_6 + 1/250$ for these six gaps.

Table 1: Exact small-gap Rayleigh comparisons.

g	N/D	$\Delta(N, D)$	$N/D - T$
1	812/97	5598897096603523	5598897096603523/1212500000000000
2	866/109	484843129173031	484843129173031/1362500000000000
3	3114/393	702232566651387	234077522217129/1637500000000000
5	764/96	587568260556064	18361508142377/375000000000000
7	768/97	98897096603523	98897096603523/1212500000000000
8	19672/2487	174815250030533	174815250030533/3108750000000000

2.3 One witness for every $g \geq 9$

On $I = [-2, 11]$, use the fixed vector

$$v_* = (4, 0, 7, 8, 8, 9, 1, 7, 3, 6, 1, 4, 1, 2), \quad \|v_*\|^2 = 391. \quad (42)$$

Only coefficients τ_i with $-4 \leq i \leq 11$ enter its nonzero image. The right defect can therefore affect the calculation only for $g = 9$ or $g = 10$; all $g \geq 11$ have the same local coefficient word. On $J = [-4, 13]$, direct evaluation gives

$$A_9 v_* = (-4, 4, 7, 3, 20, 24, 23, 24, 5, 19, 11, 15, 6, 10, 5, 5, 3, -2), \quad (43)$$

$$A_{10} v_* = (-4, 4, 7, 3, 20, 24, 23, 24, 5, 19, 11, 15, 6, 10, 5, 5, 1, -2), \quad (44)$$

$$A_g v_* = (-4, 4, 7, 3, 20, 24, 23, 24, 5, 19, 11, 15, 6, 10, 5, 5, 1, 2), \quad (g \geq 11). \quad (45)$$

Moreover,

$$\|A_9 v_*\|^2 = 3102, \quad \|A_{10} v_*\|^2 = 3094, \quad \|A_g v_*\|^2 = 3094 \quad (g \geq 11).$$

Hence every $g \geq 9$ satisfies

$$\sup \text{spec}(H_g) \geq \frac{3094}{391} = \frac{182}{23}. \quad (46)$$

The final exact comparison is

$$\Delta(182, 23) = 10563229091557 > 0, \quad \frac{182}{23} - T = \frac{10563229091557}{287500000000000} > 0. \quad (47)$$

Finite propagation, rather than sampling, is what reduces the infinite tail of gap lengths to the three images (43)–(45).

2.4 Completion of the single-gap hierarchy

For $g = 4$, the defect lattice is $4\mathbb{Z}$, so the operator is the period-eight reference bulk and

$$\sup \text{spec}(H_4) = \eta < c_6. \quad (48)$$

For $g = 6$, the global interface theorem gives

$$\sup \text{spec}(H_6) = c_6, \quad \dim \ker(H_6 - c_6) = 2. \quad (49)$$

The six small vectors and the fixed vector v_* exhaust every positive gap outside $\{4, 6\}$. Equation (15) transfers the conclusion between the two lifts, and reflection transfers it between the two orientations. The smallest printed margin is the $g = 8$ value

$$\frac{174815250030533}{310875000000000000} > 0.$$

We have proved the following strict statement.

Theorem 2.1 (Uniform abnormal single-gap separation). *For either lift and either interface orientation,*

$$\sup \text{spec}(H_6) = c_6, \quad \dim \ker(H_6 - c_6) = 2.$$

For every positive integer $g \notin \{4, 6\}$,

$$\sup \text{spec}(H_g) > c_6 + \frac{1}{250}.$$

Thus G6 is the unique minimizer among abnormal positive single gaps. The statement does not compare G6 with arbitrary multi-gap or finite-core interfaces.

3 Rational Enclosures for the G6 Atlas

This section records the outward rational enclosures whose signs are used in Appendix A of the article. They are separated from the argument so that the appendix retains the physical matching, monotonicity, and chart-completeness logic without becoming a raw certificate listing.

For

$$a_6 = \frac{7905369311620327}{10^{15}}, \quad b_6 = \frac{988171163952541}{125000000000000}, \quad J_6 = [a_6, b_6],$$

the normalized Evans determinant satisfies

$$\begin{aligned} -\frac{113736338185011273961051}{10^{30}} &\leq \mathcal{E}_6(a_6) \leq -\frac{2274726763700225479221}{2 \cdot 10^{28}} < 0, \\ 0 < \frac{4691532211683817938621607}{5 \cdot 10^{29}} &\leq \mathcal{E}_6(b_6) \leq \frac{1876612884673527175448643}{2 \cdot 10^{29}}, \\ L_D &= \frac{4748398569130503440680129312442433244027}{5 \cdot 10^{29}}, \\ U_D &= \frac{1187100548105535925201118811466127038887}{1250000000000000000000000000000}, \\ 0 < L_D &\leq \mathcal{E}'_6(y) \leq U_D \quad (y \in J_6). \end{aligned}$$

On the middle Grassmann chart,

$$\begin{aligned} 0 &< \frac{37215527374634454788819933641363}{2500000000000000000000000000000} \leq p_{01}^R, \\ p_{02}^L &\leq -\frac{24936928968874033994088618671}{10^{30}} < 0. \end{aligned}$$

For the two nonphysical resultant intervals I_1, I_2 , direct substitution into the unsquared matching determinant gives the following bounds:

$$\begin{aligned} L_1 &= -\frac{8362762401395322424628954595615433141079}{5 \cdot 10^{29}}, \\ U_1 &= -\frac{8362759240435040227417741939929556830821}{5 \cdot 10^{29}}, \\ L_2 &= -\frac{9646322659355383800530837161791302703671}{5 \cdot 10^{29}}, \\ U_2 &= -\frac{19292398194159317615673354336232538613017}{10^{30}}, \\ L_j &\leq \mathcal{E}_6(y) \leq U_j < 0 \quad (y \in I_j, j = 1, 2). \end{aligned}$$

These rows are reproduced from the exact interval certificates; no rounded decimal is used in the sign decisions.

4 Reproducibility Record

4.1 Source states and evidence boundary

Repository identifiers, source hashes, artifact paths, and the verification entry-point name are omitted from this review copy. They are included in the identified supplement and will be bound to the blinded review archive.

The checker chain re-evaluates finite mathematical contracts. Two coordinate implementations separately audit the global unsquared exclusions, whereas some transfer and elimination helpers are shared. The machine-readable manifest discloses this boundary family by family. A successful checker is evidence only together with the analytic reductions in the article and this supplement.

4.2 Certificate families

Family	Logical role
G6 matching and edge	Physical matching, elimination, atlas completeness, and the global one-interface edge.
Separated interfaces	The supplementary exact $2r$ -cluster theorem; not used in the classification.
Single-gap hierarchy	Finite propagation and strict integer Rayleigh comparisons for all abnormal positive single gaps.
Orders 8–30	Complete switching-class lower bounds at the initial equality orders.
Order 32	First strict counterexample and independent signed-matrix reconstruction.
Orders 34–46	Finite-state completeness, the recovery orders, and the isolated failure at 40.
Orders 48–238	The finite bridge from the uninterrupted failure onset to the analytic tail.

Table 2: Proof-certificate families and their role in the article.

The blinded review archive contains a machine-readable manifest mapping each family to certificate digests, producer and verifier roles, its independence statement, and the hardware profile.

4.3 One-command verification

From the root of a clean checkout, the identified package is verified by the single entry point documented in its blinded archive. The entry point first audits the manuscript, bibliography, anonymity, certificate manifest, digests, and frozen historical source trees. It then runs all proof-grade verifiers and the focused tamper-test suites. The exact tasks are CPU-only; no GPU library is required. Python 3.11 or later and 8 GiB of memory suffice, with 16 GiB recommended for comfortable parallel use. The measured reference runtime for the final checkpoint is recorded in the Task 59 handoff rather than hard-coded into this document.

4.4 Scope

The separated-interface theorem is restricted to $r \in \{1, 2, 3\}$ and $D \geq 1040$. It counts eigenvalues with multiplicity and proves neither individual simplicity nor a sign for the leading interaction. The single-gap theorem concerns positive single gaps only. Neither result classifies arbitrary multi-interface minimizers.